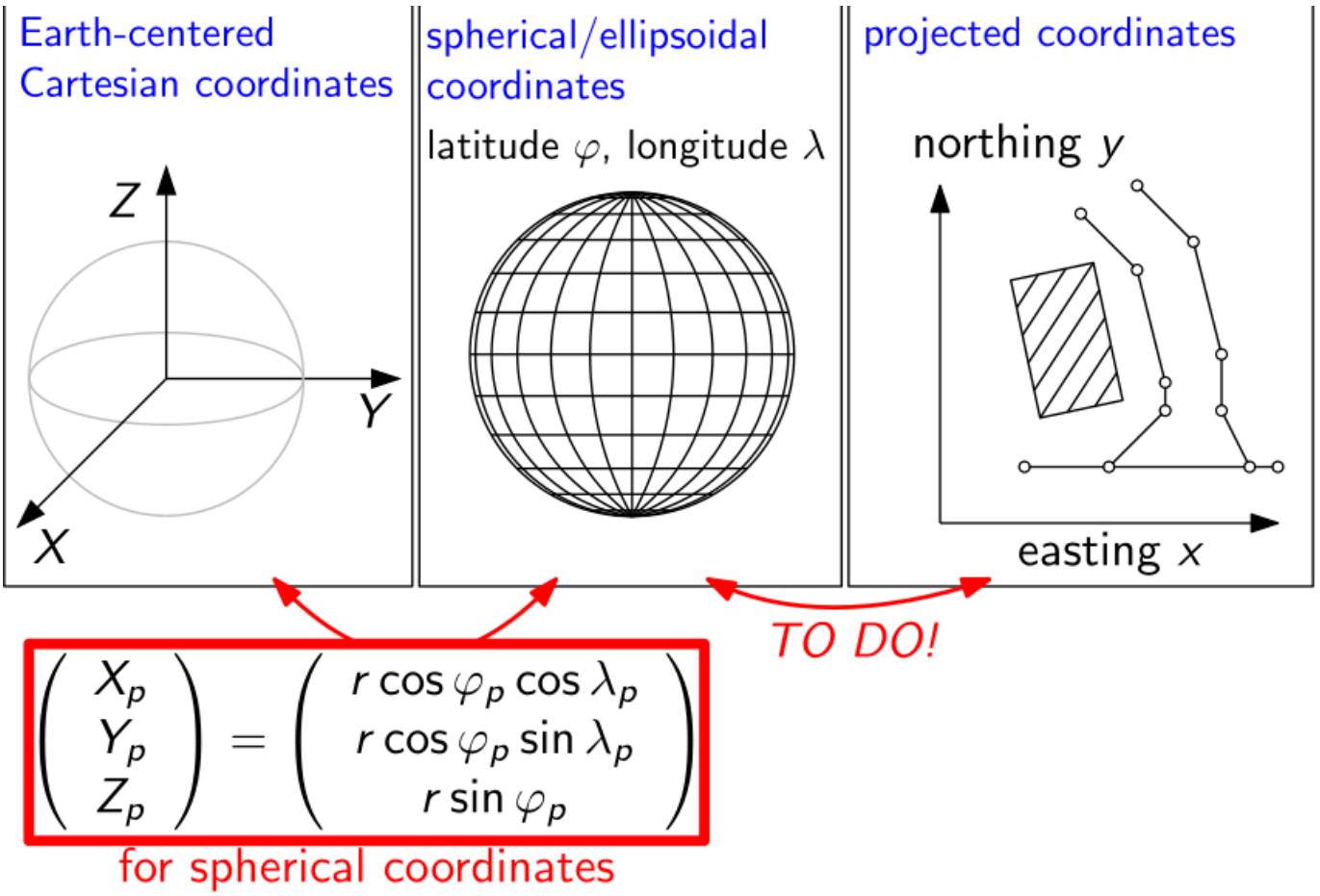
L7: Map Projections



Cylincrical Projection

- e.g. Plate Carrée
- For **all** cylincrical projections in normal position:
  - Lines of equal latitude are mapped to horizontal lines
  - Meridians are mapped to vertical lines
  - Equator scaled with constant factor (often 1)
- Distances between lines of equal latitue can be non-uniform
- For *transverse* cylindrical projections, consider one meridian as the Earth's equator
  - Mercator projection

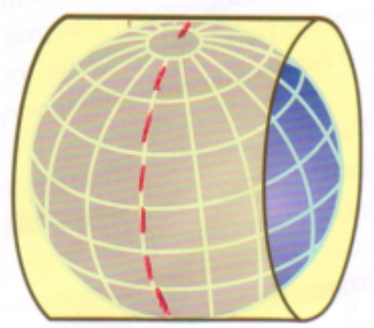
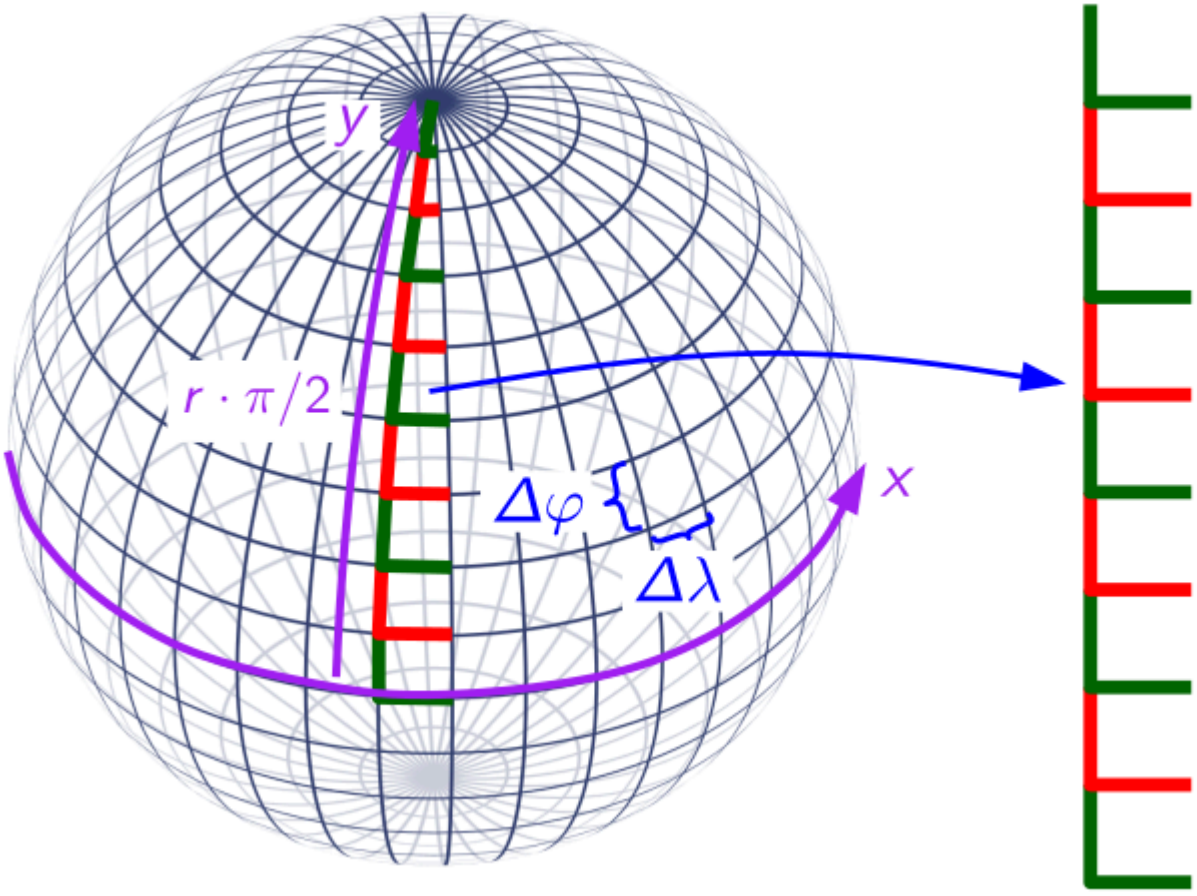
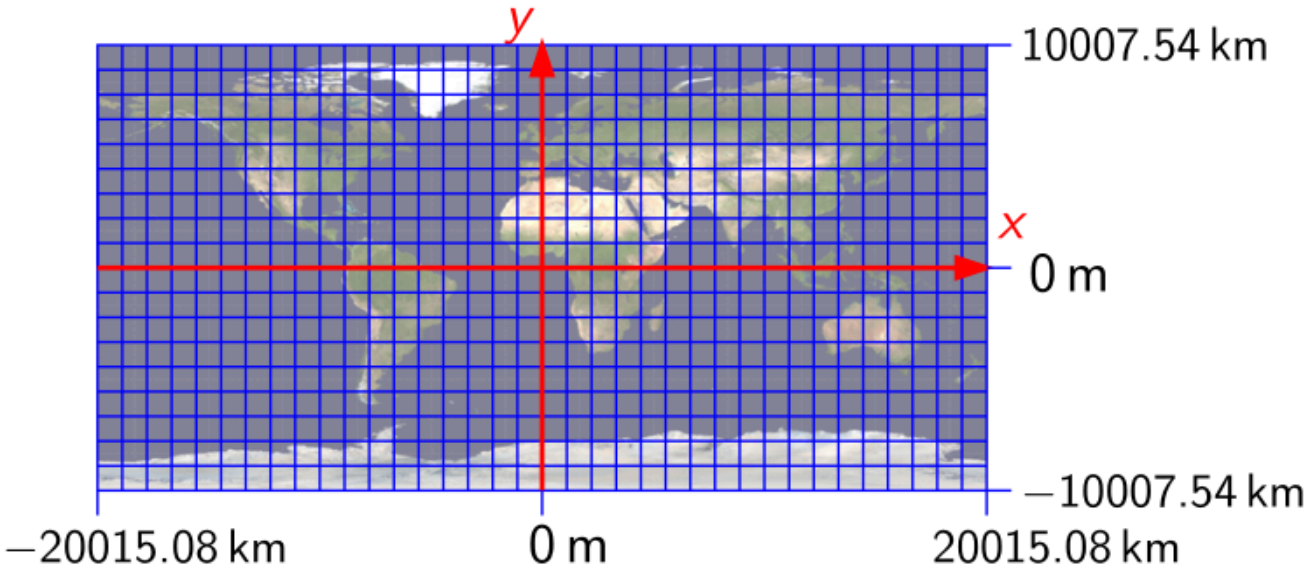
- 

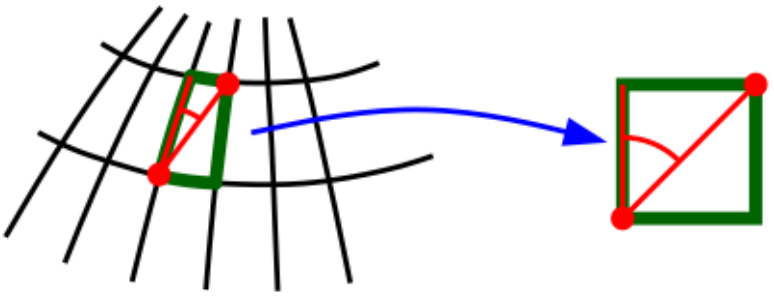
Plate Carrée Projection

- Map areas between lines of constant latitude/longitude to **squares of constant size** (assuming $\Delta\varphi = \Delta\lambda$)
- **Preserve lengths of equator and meridians**
- 

- **Mapping Function**:
  - Easting $x = \lambda_{[deg]} \cdot \frac{\pi}{180^\circ} \cdot r$
  - Northing$y = \varphi[deg] \cdot \frac{\pi}{180^\circ} \cdot r$
  - with r = Earth radius

- $\begin{pmatrix} x_p \\ y_p \end{pmatrix} = \begin{pmatrix} r\lambda_p \\ r\lambda_p \end{pmatrix}$





Disadvantages

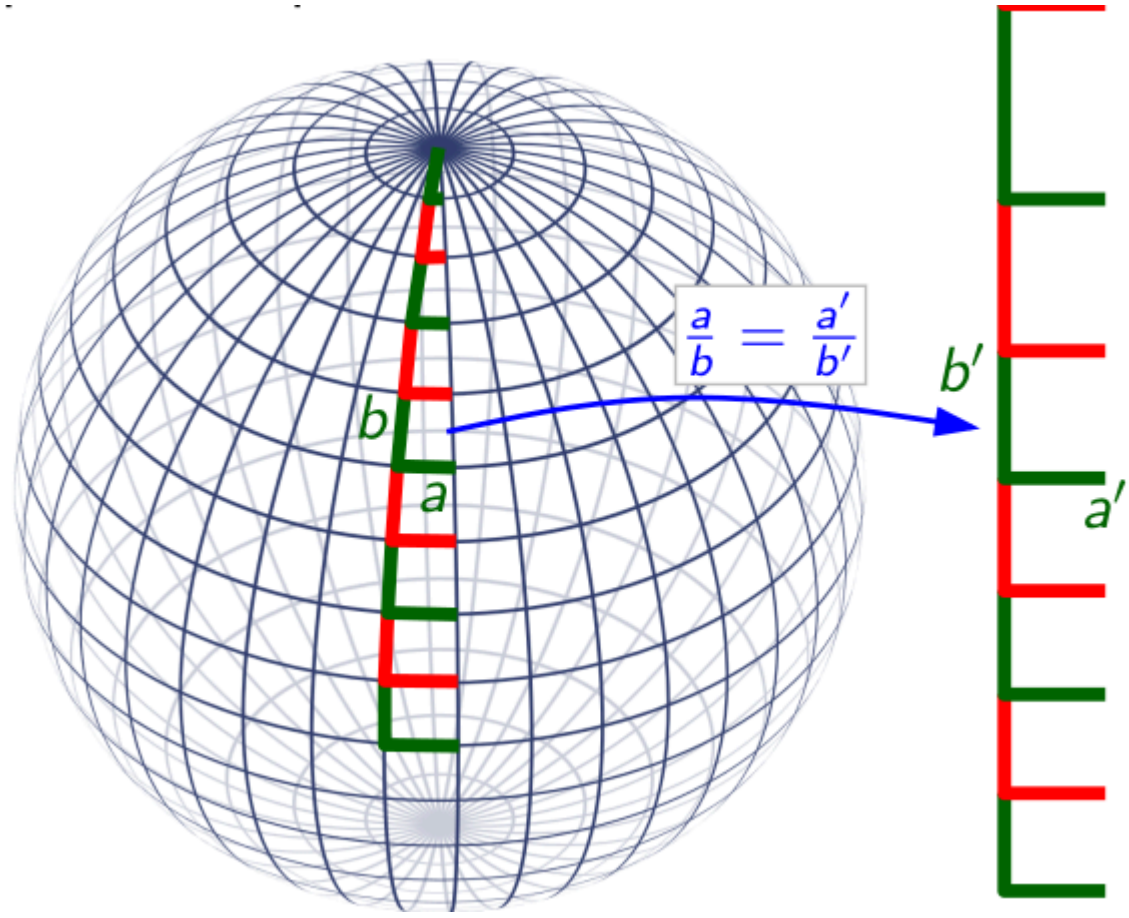
- **Shapes** get "Squeezed", i.e. aspect ratios change
- **Angles** change
- **Areas** Change

Advantages

- **Lengths** of **equator** and **meridians** are preserved

Mercator Projection

- Map areas between lines of constant latitude/longitude to rectangles that preserve aspect ratios
- Here shown for $\Delta\lambda = \Delta\varphi = 10^\circ$
- For exact construction: Choose $\Delta\lambda, \Delta\varphi$ infinitely small



- Mapping Function
  - $x_p = r\lambda_p$
  - $y_p = r \ln \tan(\frac{\pi}{4} + \frac{\varphi_p}{2})$
    - r = Earths's radius

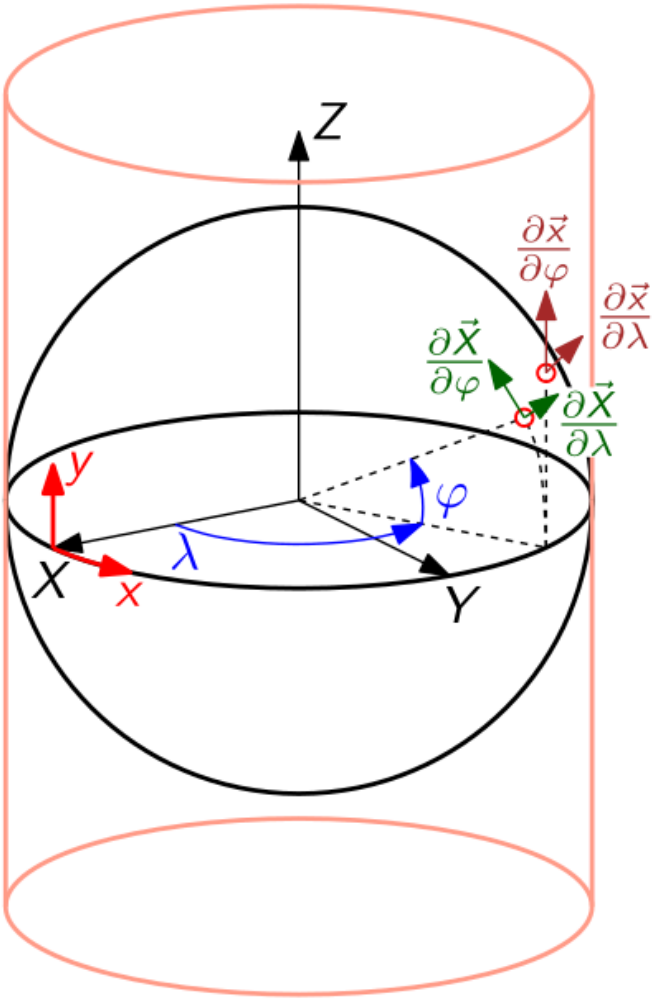
Disadvantages

- Areas are distored (esp. close to poles)
- Lengths of meridians are not preserved

Advantages

- Aspect ratios are preserved
- (->) Angles are preserved
- Mercator projection is conformal

Mercator Mapping function derivation



Requirement:
same aspect ratios

$$\frac{\left|\frac{\partial \vec{x}}{\partial \varphi}\right|}{\left|\frac{\partial \vec{x}}{\partial \lambda}\right|} = \frac{\left|\frac{\partial \vec{X}}{\partial \varphi}\right|}{\left|\frac{\partial \vec{X}}{\partial \lambda}\right|}$$

$$\Leftrightarrow \frac{\left(\frac{\partial \vec{x}}{\partial \varphi}\right)^2}{\left(\frac{\partial \vec{x}}{\partial \lambda}\right)^2} = \frac{\left(\frac{\partial \vec{X}}{\partial \varphi}\right)^2}{\left(\frac{\partial \vec{X}}{\partial \lambda}\right)^2}$$

$$\Leftrightarrow \frac{\left(f'(\varphi)\right)^2}{r^2} = \frac{r^2}{r^2 \cos^2 \varphi}$$

$$\Leftrightarrow f'(\varphi) = \frac{r}{\cos \varphi}$$

$$\underset{f(0)=0}{\Leftrightarrow} f(\varphi) = r \ln \tan\left(\frac{\pi}{4} + \frac{\varphi}{2}\right)$$

$$\left(\frac{\partial \vec{x}}{\partial \varphi}\right)^2 = \begin{pmatrix} 0 \\ f'(\varphi) \end{pmatrix}^2 = (f'(\varphi))^2$$

$$\left(\frac{\partial \vec{x}}{\partial \lambda}\right)^2 = \begin{pmatrix} r \\ 0 \end{pmatrix}^2 = r^2$$

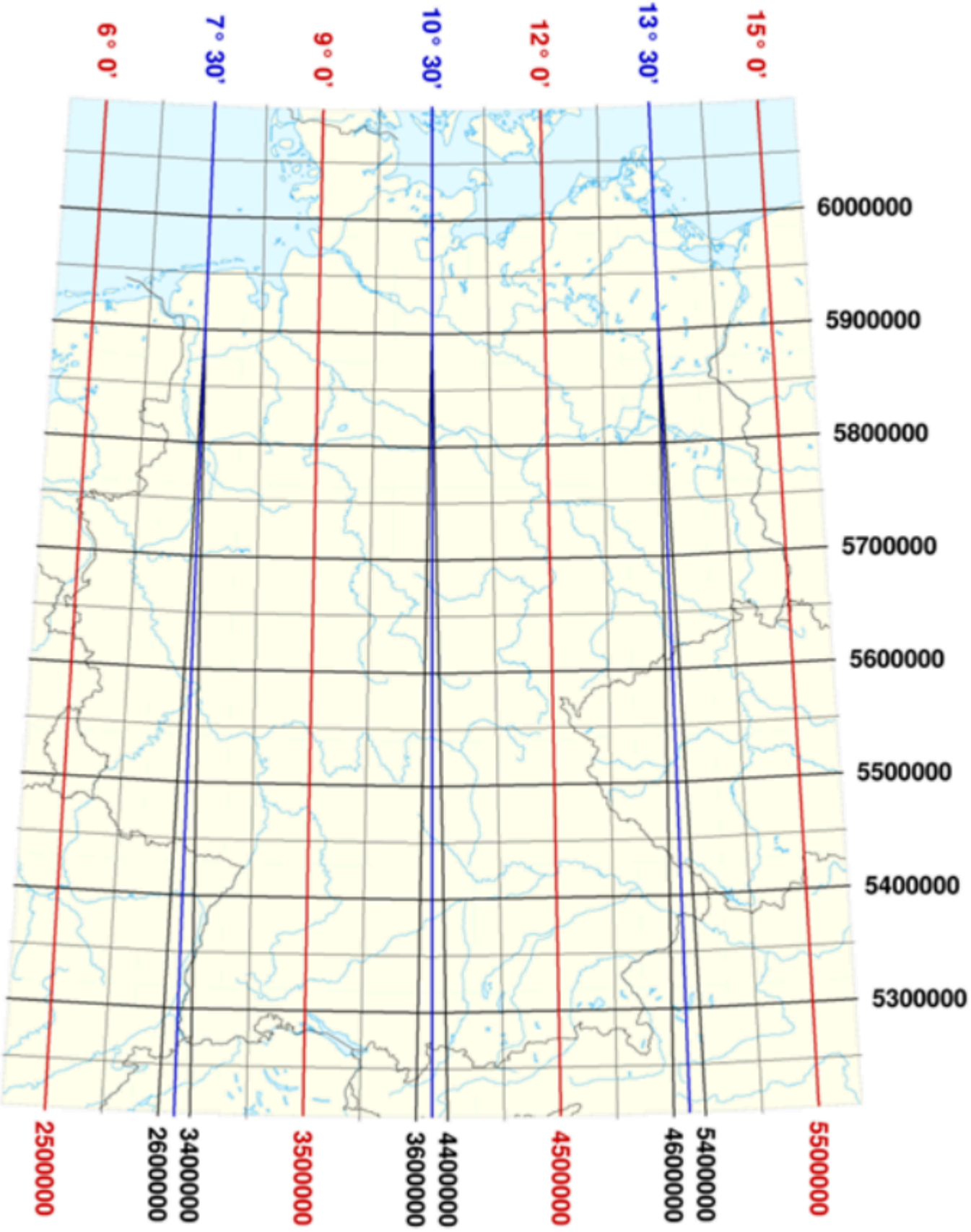
$$\left(\frac{\partial \vec{X}}{\partial \varphi}\right)^2 = \begin{pmatrix} -r\sin\varphi\cos\lambda \\ -r\sin\varphi\sin\lambda \\ r\cos\varphi \end{pmatrix}^2 = r^2$$

$$\left(\frac{\partial \vec{X}}{\partial \lambda}\right)^2 = \begin{pmatrix} -r\cos\varphi\sin\lambda \\ r\cos\varphi\cos\lambda \\ 0 \end{pmatrix}^2 = r^2\cos^2\varphi$$

with

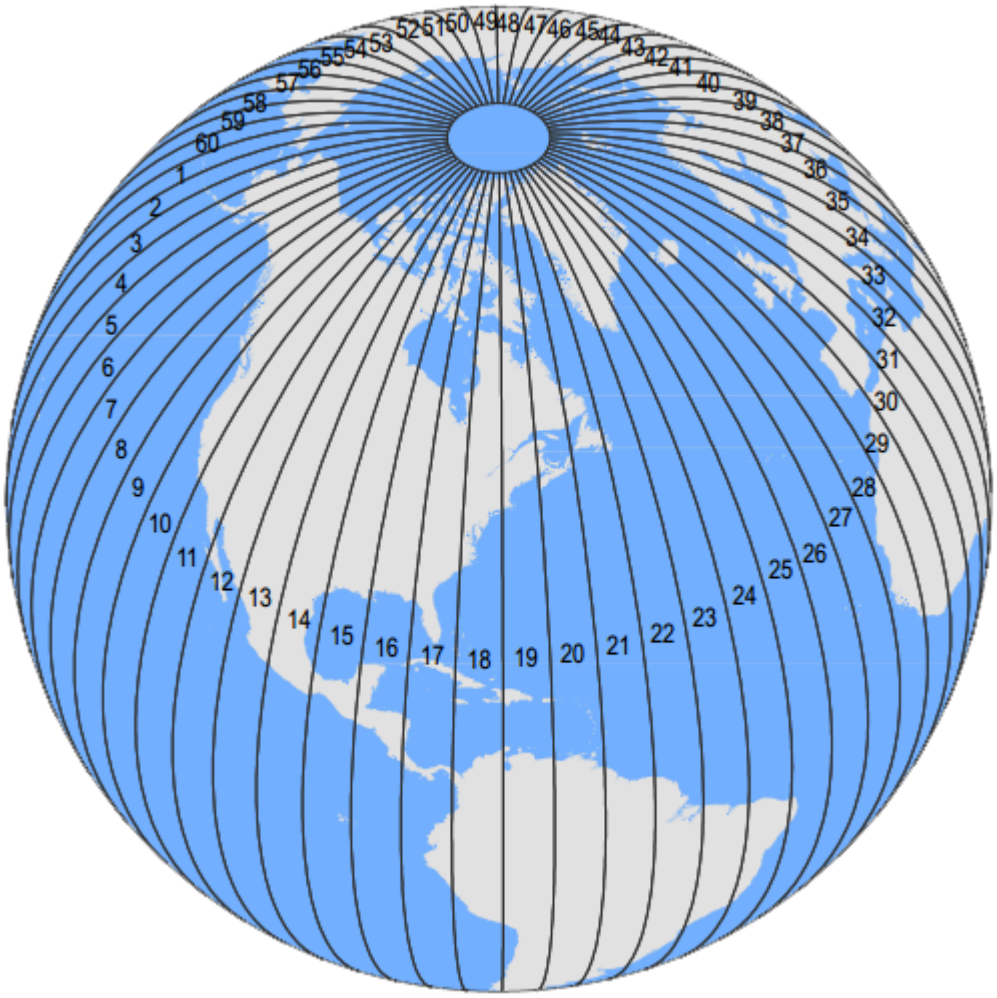
Gauß Krüger projection

- Transversed Cylincrical Projection
- Earth subdivision into 3° width zones
- For different zones, differenz projections
  - Central meridian = center of zone
- Western German system (replaced by UTM)



Universal Transverse Mercator - UTM

- Similar to Gauß-Krüger
- Width of zon 6°
- Scale=0.9996



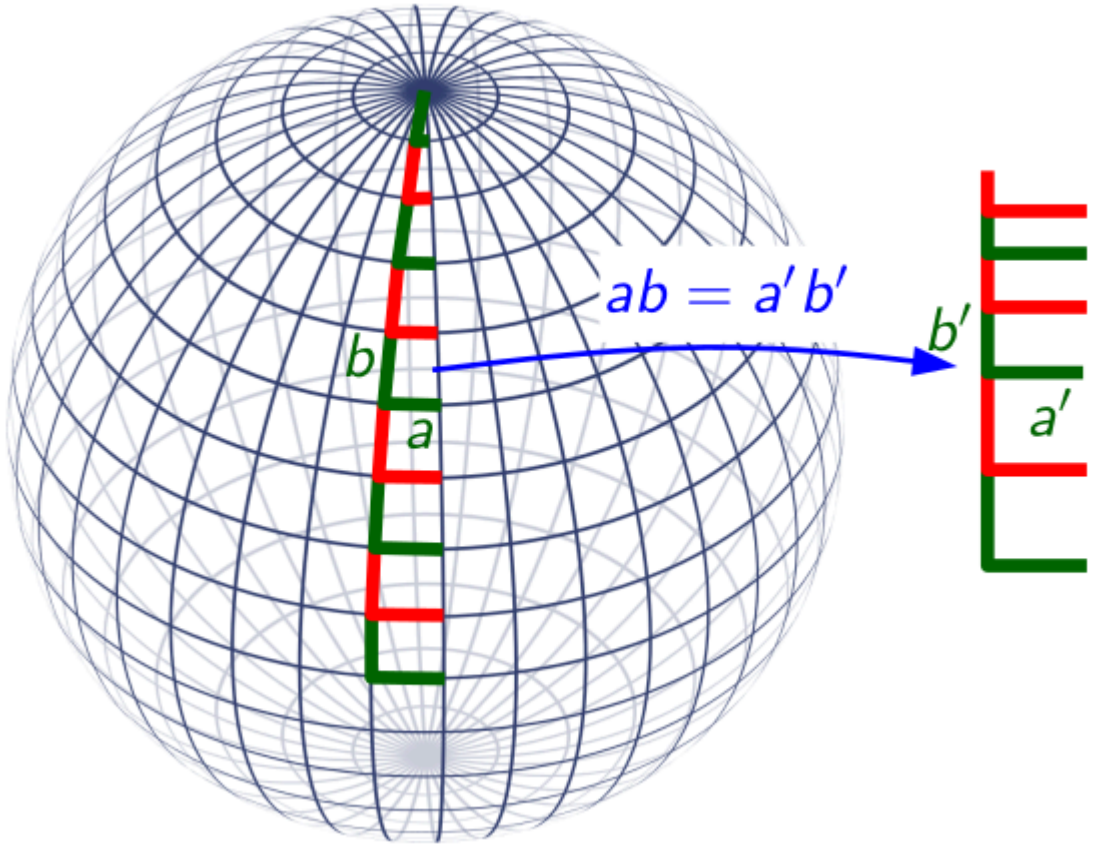
- **Mapping function**:
  - $x_p = r\lambda_p$
  - $y_p = r \ln \tan\left(\frac{\pi}{4} + \frac{\varphi_p}{2}\right)$

Conclusion

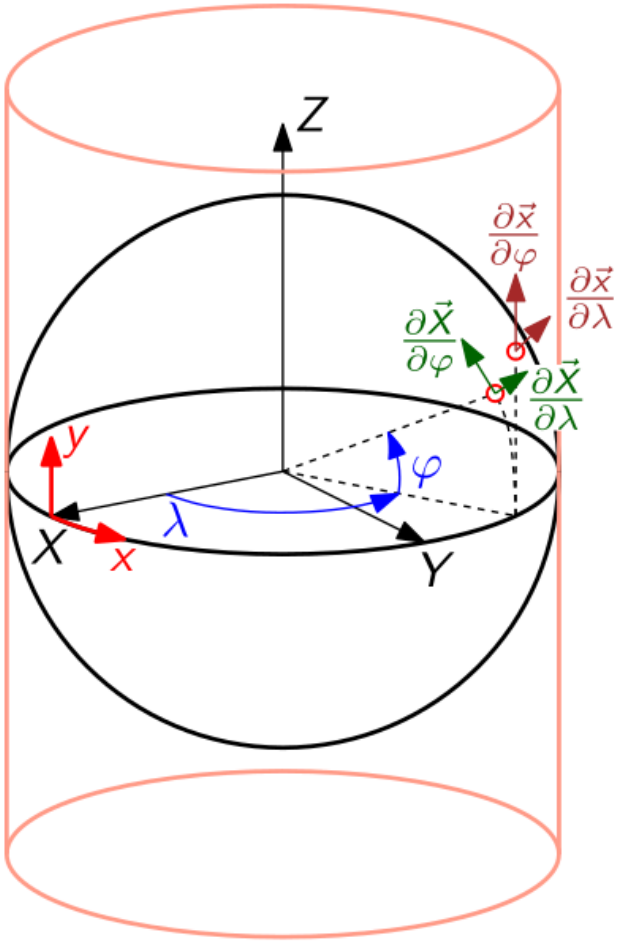
- Mercator projection not appropriate for world maps
- Frequently usedf for smaller zones
  - Close to equator or
  - Close to a central meridian (transverse version)

Cylindrical Equal-Area Projection

- Map areas betewen lines of constant latitude/longitude to rectangels of the same sizes
- Here shown for $\Delta\lambda = \Delta\varphi = 10^\circ$
- For exact construction: $\Delta\lambda, \Delta\varphi$ infinitely small



Cylindrical Equal-area Projection – Maths



Requirement:
same sizes

$$\left|\frac{\partial \vec{x}}{\partial \varphi}\right| \cdot \left|\frac{\partial \vec{x}}{\partial \lambda}\right| = \left|\frac{\partial \vec{X}}{\partial \varphi}\right| \cdot \left|\frac{\partial \vec{X}}{\partial \lambda}\right|$$

$$\Leftrightarrow \left(\frac{\partial \vec{x}}{\partial \varphi}\right)^2 \left(\frac{\partial \vec{x}}{\partial \lambda}\right)^2 = \left(\frac{\partial \vec{X}}{\partial \varphi}\right)^2 \left(\frac{\partial \vec{X}}{\partial \lambda}\right)^2$$

$$\Leftrightarrow (f'(\varphi))^2 r^2 = r^4 \cos^2 \varphi$$

$$\Leftrightarrow f'(\varphi) = r \cos \varphi$$

$$\Leftrightarrow f(\varphi) = r \sin \varphi$$

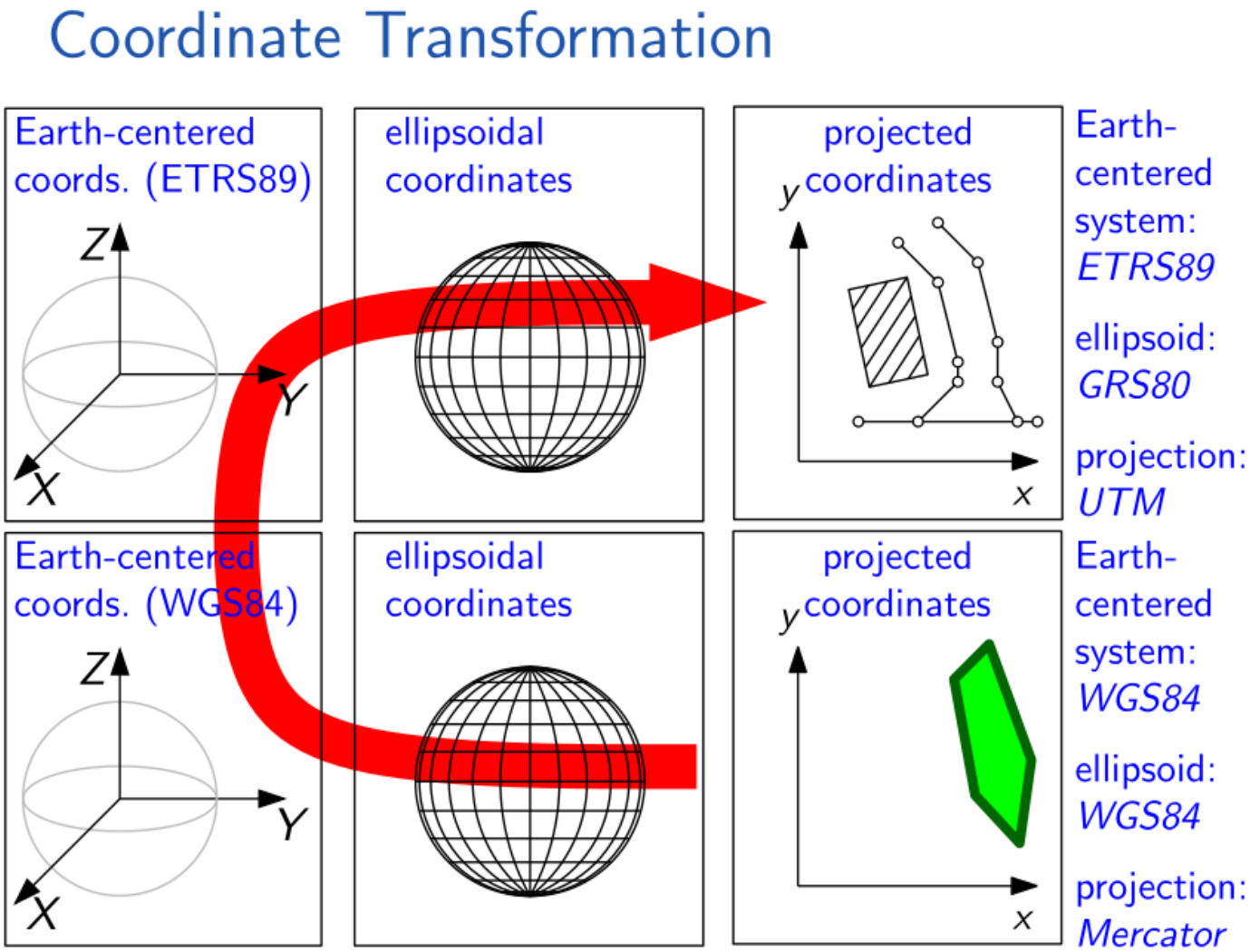
$f(0) = 0$

- **Mapping function**:
  - $x_p = r\lambda_p$

- $y_p = r \sin \varphi_p$
  - r = Earth's radius

Disadvantages

- Cylindrical equal-area projection leads to large distortions of aspect rations and angles

More Map Projections

- For whole World: Compromise between area distortion and angle distortion
  - e.g. Robinson projection
- No map projection preserves both areas and angles!